



Green University of Bangladesh

*Department of Computer Science and Engineering (CSE)
Semester: (Spring, Year: 2024), B.Sc. in CSE (EVE)*

Simple Point of Sale System Using Java

Course Title: Object Oriented Programming Lab

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<u>Lab Project Status</u>	
Marks:	Signature:
Comments:	Date:

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Chapter 1

Introduction

1.1 Overview

The Java Cafe POS System project aims to develop a comprehensive point-of-sale (POS) solution specifically designed for cafes. The system will streamline various tasks in running the café, including order management, inventory tracking, and sales reporting, ultimately increasing efficiency and customer service.

1.2 Motivation

The decision to launch the Java Cafe POS System project is driven by the increasing demand for high-performance POS solutions in the food service industry, especially in the cafe industry.

Traditional POS systems often lack the necessary flexibility to meet the specific needs of cafes, leading to inefficiency and operational challenges. [1].

1.3 Problem Definition

1.3.1 Problem Statement

The main challenge that the Java Cafe POS system seeks to address is the lack of a POS solution modified to the cafe. Existing systems tend to be generic and not optimized for the specific business processes and needs of the cafe industry , and difficulty in accurately monitoring inventory and sales data.

1.3.2 Complex Engineering Problem

The following Table 1.1 must be completed according to your above discussion in detail. The column on the right side should be filled only on the attributes you have chosen to be touched by your own project.

Table 1.1: complex engineering problems are addressed:

Name of the P Attributes	Explain how to address
P1: Depth of knowledge required	Developing the POS system requires a deep understanding of Java programming, GUI development with Swing, and basic principles of software engineering.
P2: Depth of analysis required	In-depth analysis is needed to optimize the user interface, ensure proper database management, and seamlessly integrate applications.
P3: Interdependence	The components of the system are highly interdependent; For example, the order management module relies on accurate inventory data, and the billing module relies on order and payment information.

1.4 Design Goals/Objectives

Customization: Develop a flexible and customized POS system that can allow various menu items with different quantities and prices.

User-Friendly Interface: Design an built-in user interface that simplifies the process of adding items to the order list and calculating the total bill.

Chapter 2

Design/Development/Implementation of the Project

2.1 Introduction

The Java Cafe POS System project involves the design, development, and implementation of a point-of-sale solution specifically for cafes. The goal is to create a user-friendly, efficient, and scalable POS system that addresses the unique needs of cafe operations, including order management.

2.2 Project Details

The system of the Java Cafe POS System is designed to be modular and scalable, ensuring that various components can be developed, tested, and maintained independently. The primary components of the system include:

1. **User Interface (UI):** The UI is developed using Java Swing components such as JFrame, JButton, JTable, JTextArea, and JTextField. The interface is designed to be intuitive and user-friendly, allowing café staff to quickly add items to orders, view the current order list, and process payments.
 - **Order Table:** Displays the current order list, including item ID, name, quantity, and price.
 - **Buttons:** Provide functionality for adding items to the order, removing items, and processing payments.
 - **Text Areas and Fields:** Used for displaying the bill and inputting payment amounts.
2. **Order Management Module:** Handles the addition, update, and removal of items from the order list. It ensures that the order details are accurately maintained and updated in real-time.

3. **Billing Module:** Generates detailed bills, calculates totals, and processes payments. It formats the bill for display and printing.
4. **Payment Processing Module:** Manages the calculation of the total amount payable, processes the payment input, and computes the balance to be returned to the customer.

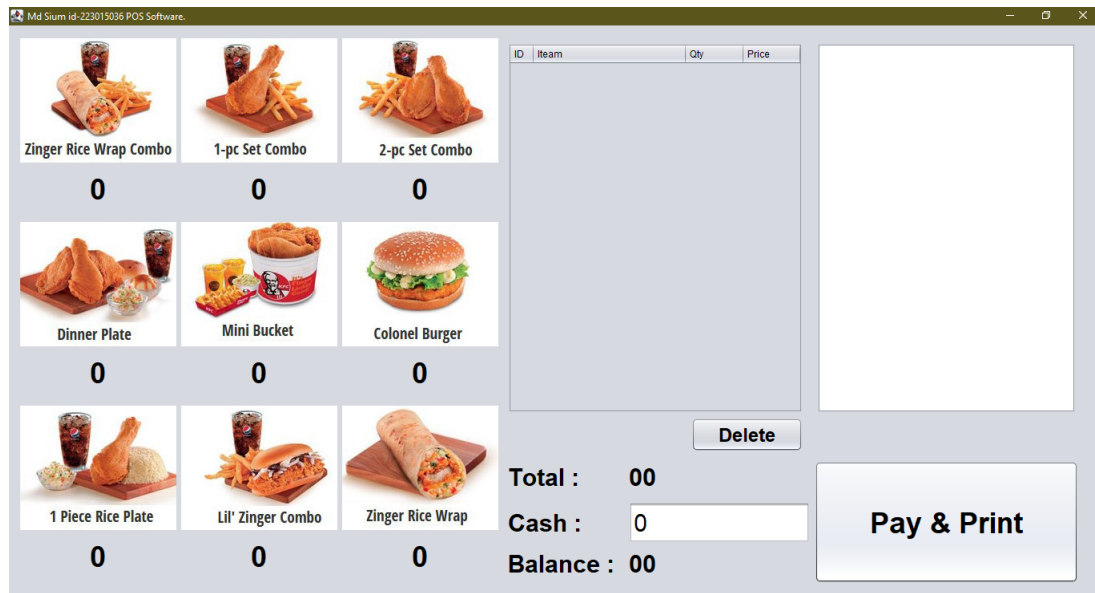


Figure 2.1: UI

You can fix the height, width, position, etc., of the figure accordingly.

2.3 Tools and Libraries

1. IntelliJ IDEA .
2. GUI Framework

Java Swing: Swing components such as JFrame, JButton, JTable, JTextArea, and JTextField are used extensively to create the graphical user interface of the POS system.

Chapter 3

Performance Evaluation

3.1 Results Analysis/Testing

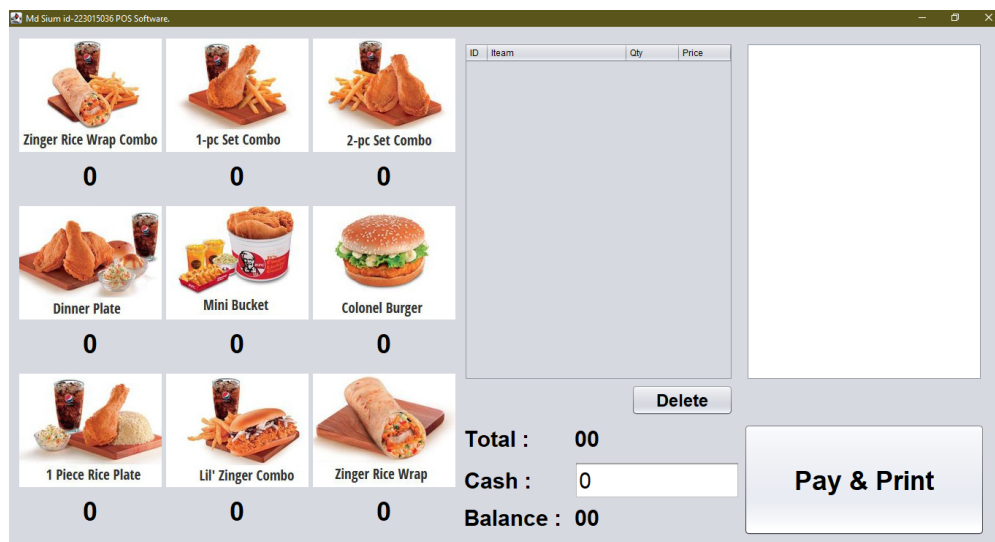


Figure 3.1: A graphical result of project

Md Sium id-223015036 POS Software.





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ID	Item	Qty	Price
1	ZRW Combo	1	02.99
4	Dinner Plate	1	09.29
5	Mini Bucket	1	08.19
6	Col Burger	1	03.99
7	1pc Rice Plate	1	12.99
8	Lil' Combo	1	13.99
9	Rice Wrap	1	04.99
2	1pc Combo	3	07.77
3	2pc Combo	1	07.99

GREEN CAFE
PURBACHAL AMERICAN CITY,
Dhaka, bangladesh
0172014617,

Item	Qty	Price
ZRW Combo	1	02.99
Dinner Plate	1	09.29
Mini Bucket	1	08.19
Col Burger	1	03.99
1pc Rice Plate	1	12.99
Lil' Combo	1	13.99
Rice Wrap	1	04.99
1pc Combo	3	07.77
2pc Combo	1	07.99

Sub Total : 72.19
Cash : 73
Balance : 00.81

Thanks For Your Business...!

Software by Md Sium

Delete

Total : 72.19
Cash : 73
Balance : 00.81

Pay & Print

Figure 3.2: Order Item

GREEN CAFE
PURBACHAL AMERICAN CITY,
Dhaka, bangladesh
0172014617,

Item	Qty	Price
ZRW Combo	1	02.99
Dinner Plate	1	09.29
Mini Bucket	1	08.19
Col Burger	1	03.99
1pc Rice Plate	1	12.99
Lil' Combo	1	13.99
Rice Wrap	1	04.99
1pc Combo	3	07.77
2pc Combo	1	07.99

Sub Total : 72.19
Cash : 73
Balance : 00.81

Thanks For Your Business...!

Software by Md Sium

Figure 3.3: Order Print

Chapter 4

Conclusion

4.1 Discussion

Java Cafe POS System project, detailing its design, development, and implementation phases. The system was designed to address the specific needs of cafes, focusing on order management, billing, and seamless payment processing. Through the utilization of Java Swing for the user interface and we have successfully created a prototype that demonstrates the core features of a POS system tailored for small businesses.

4.2 Limitations

Despite the progress made, the Java Cafe POS System has several limitations that warrant consideration. Firstly, the current prototype lacks integration with a robust database management system for persistent data storage, which limits its ability to manage customer information effectively. Secondly, while the user interface is functional.

4.3 Scope of Future Work

several avenues for future work and project extension to enhance the Java Cafe POS System:

Database Integration: Implement a robust database solution MySQL to manage, customer data, and transaction histories securely and efficiently.

References

- [1] Omid C Farokhzad and Robert Langer. Impact of nanotechnology on drug delivery. *ACS nano*, 3(1):16–20, 2009.